LLM Fundamentals

↓

Large Language Model.

Gemini
Gpt
Claude
Bert
≡

→ Tokenization

→ Transformer ⇒ Self attention

↳ Attention is all you need.

→ Context - Window

→ Temperature

Data

↳ Huge amount of data

→ Books

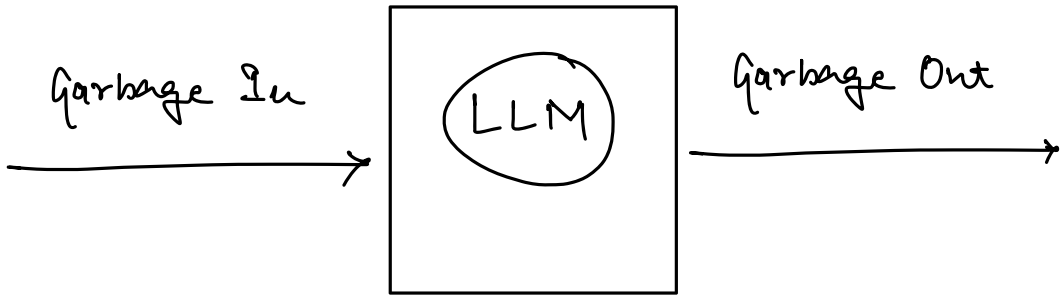
→ Research Papers

→ Websites

→ Wikipedia

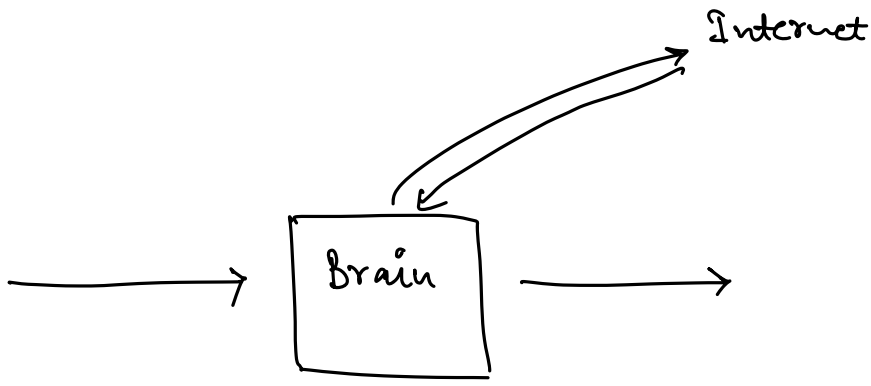
→ blogs.

Input → [LLM.] → Generate O/P.

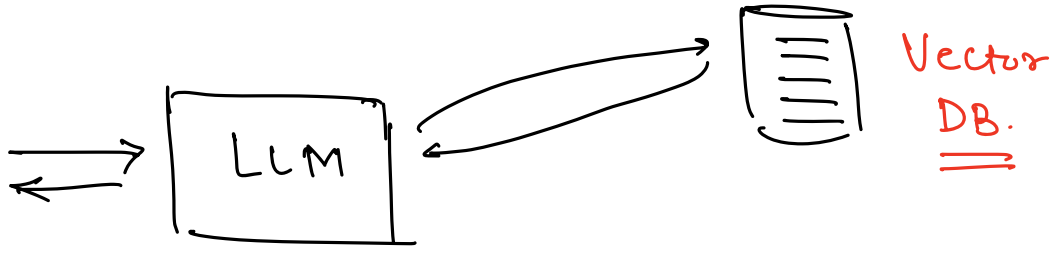


→ Cut-off

Hallucination



ChatBot for Masai



#

Hi, How are you?

↓

GPT

↓

Tokenization → Breaking down the sentence | text into small units

- words
- subwords
- Character

Hi, How are you ?

I love coding

↓

["I", "Love", "coding"] → Word Level Tokenization

['I', 'l', 'o', 'v', 'e', - - - - -] → Character Level Tokenization

⇒ How LLMs uses Tokenization

1. Tokenization

⇒ I love coding

↓

["I", "love", "coding"]

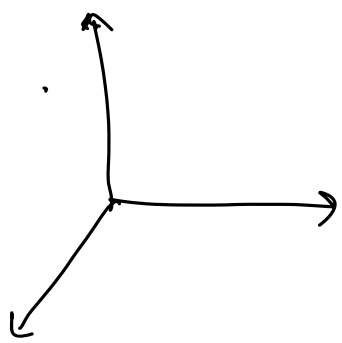
2. Convert Tokens into TokenIDs.

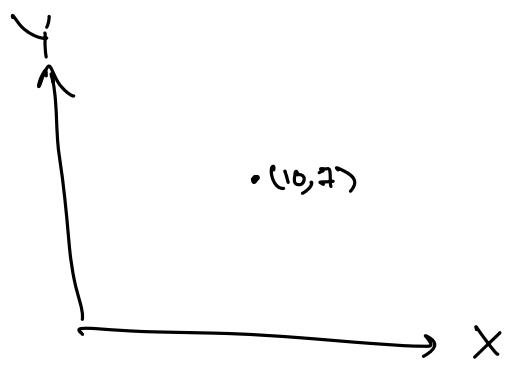
["I", "love", "coding"]

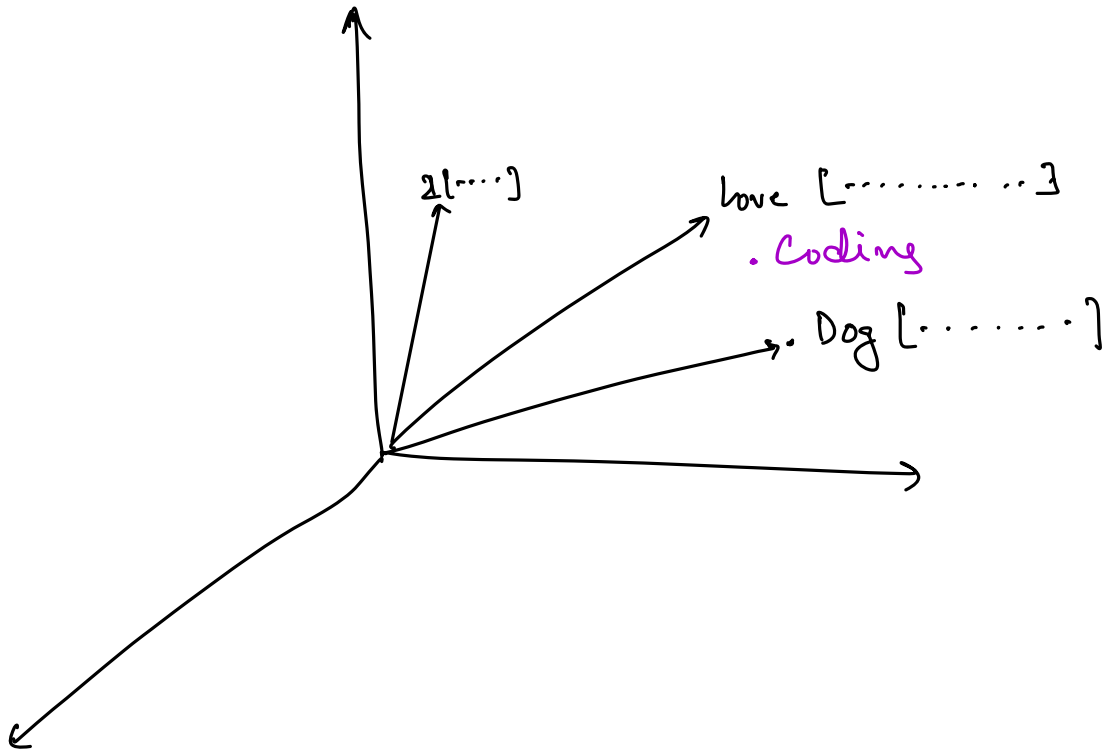
71 4165 71612

[71, 4165, 71612]

3.







Attention mechanism

→ Importance of a word in a sentence.

→ Right context of the word

→ The river bank

→ The Hdfc bank

Sanju missed century by 3 runs.

AI is the biggest revolution of 21st century

⇒ Model is now trained by predicting the next set of tokens.

⇒ Model gives the answers with high probability

Human feedback.

Parameters.

→ Numbers that our model learns during training.

Model's knowledge base.

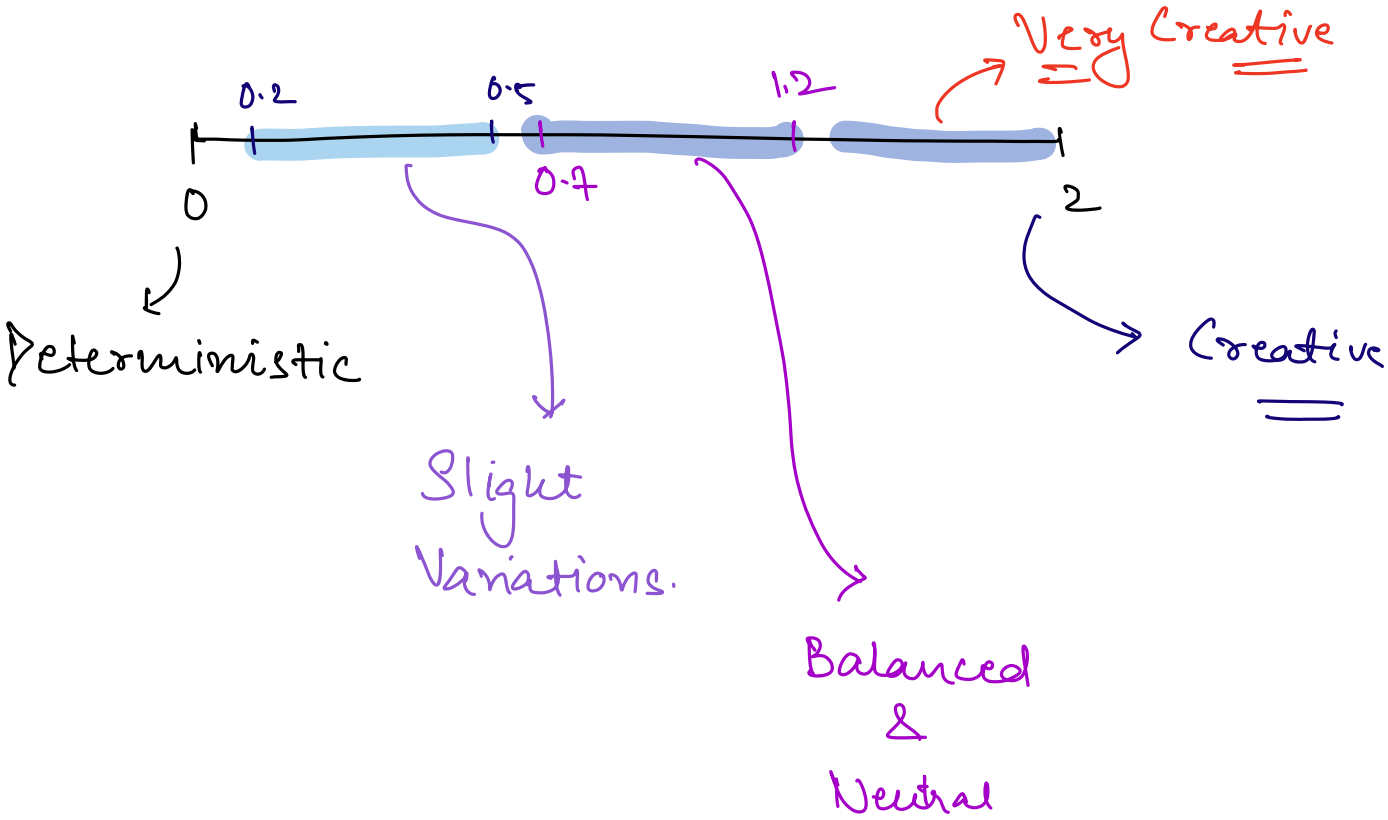
Context Window

→ Maximum amount of Tokens a LLM can look at and remember at a time to understand user's query & answer appropriately.

Temperature

→ Controls how random or deterministic the model's output is.

→ Creativity.



Sampling Methods

→ Decides which tokens to use.